

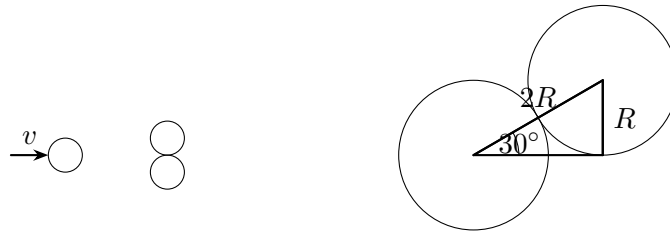
2025 F=ma Exam Review

Nathan P.

February 8, 2026

Problem 2

It's a good idea to form the triangle to determine that disk 2 and 3 would have $v \cos 30^\circ$. Set up momentum and energy conservation and solve the system; the sign of v_f is direction-sensitive.



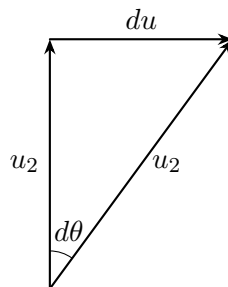
Problem 3

Key idea: it's colliding with equal masses, so it will transfer all its momentum in the component parallel to the line of action of its impulse while keeping the perpendicular/tangential component. Draw it out.

Problem 4

It's like circular motion. Use the definition of acceleration and some small-angle vector approximations: $s = r d\theta$ and $\omega = v/r$.

$$a = \frac{dv}{dt} = \frac{du}{dt} = u_2 \frac{d\theta}{dt} = u_2 \omega = \frac{u_1 u_2}{L}, \quad \omega = \frac{u_1}{L}.$$



Problem 5

Write out all force equations; solve for N_1 , N_2 , f_1 , and f_2 . Since $f \leq \mu N$, we have $\mu \leq f/N$.

Problem 6

Write it out and make sure to distinguish the height drop on the ramp and the height of the fall. The fall time $t = \sqrt{2h/g}$ is constant because the height of fall is constant. The horizontal velocity when shooting off the ramp is proportional to the square root of the height drop on the ramp: $v = \sqrt{2gh}$, and $x = vt$. So $2.5x$ means $2.5v$, which means $6.25h$ (height drop on the ramp).

Problem 7

$$K_1 + K_2 + U_g + U_s.$$

Rule out B and D because you don't subtract energy. Rule out E because if θ_1 and θ_2 were the same then it'd be

$$\frac{1}{2}m\ell^2\omega^2 + \frac{1}{2}m(2\ell\omega)^2 = \frac{5}{2}m\ell^2\omega^2,$$

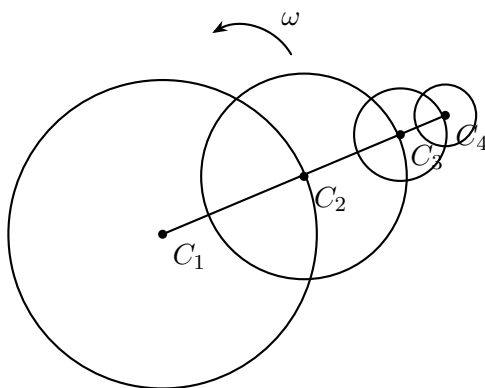
which is greater than the $\frac{1}{2} + \frac{1}{2} m\ell^2\omega^2$ in option E. Option C makes sense for being $\theta_1 - \theta_2$ because if θ was 45° that should maximize K , not make the term disappear.

Anyways, the real way to do it is $K = K_1 + K_2$. Get K_2 with $\frac{1}{2}mv_2^2$ and $v_2 = v_1 + v_{\text{rel}}$, and v_{rel} is just $\ell\omega_2$. So v_2^2 is the dot product of $v_1 + v_{\text{rel}}$ with itself:

$$v_2^2 = v_1^2 + \ell^2\omega_2^2 + 2v_1(\ell\omega_2)\cos(\theta_1 - \theta_2).$$

Problem 9

If you imagine the position vector $\mathbf{r}$ from C_1 to C_2 and from C_2 to C_3 and so on, they all spin at ω , so it'll just be a straight line spinning at ω . Don't be fooled by the answer options; make your own judgment and recognize the simplicity they're testing in these complex-looking dynamics problems.



Problem 12

My mistake was being too hasty and writing 0.5 for 50 grams instead of 0.05. Be careful with numbers. When correcting, I mistakenly put $I = \frac{1}{2}mr^2$ when for point masses it's just $I = mr^2$.

Otherwise this was a simple momentum conservation question. You could also just have used linear momentum because they are point masses.

Problem 14

It's good to recognize that it'd be harder to get a shell rolling, so that would stop slipping later. My mistake was thinking mass affected acceleration: $\mu mg = ma$, so m cancels. Frictional a is independent of mass while α depends on rotational inertia.

Problem 16

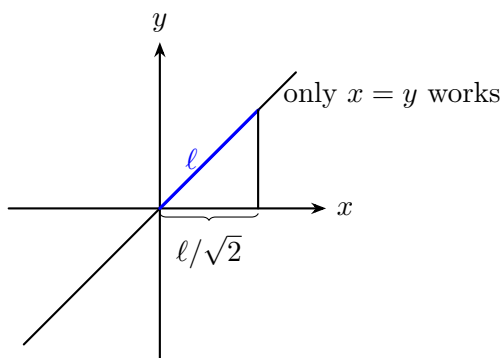
Pressure is inversely proportional to radius: $\Delta P \propto 1/R$. The smaller the bubble, the higher the pressure.

Problem 17

The fact that it's negative means the conservative force pushes it outward (the graph is a hill); regular springs have positive U . You can treat x and y directions as independent. $x^2 + y^2 (= r^2)$ means it's pushed radially outward. The force is growing, so the trajectory will approach a line.

Problem 18

x depends on y because they are multiplied. "Some direction" must be $y = x$ so that $kxy/2 = kx^2/2$, so it's stable. Pushing ℓ in the direction $y = x$ would have $x = \ell/\sqrt{2}$ and $y = \ell/\sqrt{2}$, so $xy = \ell^2/2$. So $U = k(\ell^2/2)/2$, so $k_{\text{eff}} = k/2$.



Problem 19

$P \propto E \propto mv^2$, $m \propto v$ (continuity), and given $v \propto h$: $2h$ means $2v$ means $8P$.

Problem 20

Giving it a radial velocity affects a and T negligibly.

Problem 21

It's a good idea to use the continuity equation. Careful: the question asks for the difference in the speeds.

Problem 22

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2, \quad \Delta P = P_1 - P_2 = \rho g \Delta h.$$

Solve for Δh .

Problem 23

Find an expression for the variable you're measuring, then find relative uncertainties. If a measurement has "smallest divisions" at some unit, the uncertainty is half that unit. If you take two measurements, Δx is the average of the difference of the measurements, while the relative uncertainty is Δx divided by the average of the measurements. Once you see $\Delta g/g$ is negligible, don't even bother calculating it.

Problem 25

Given $M \gg m$, it is an inertial frame traveling at constant velocity, so doing physics in that frame works. We can conserve energy to get that m ends up with $\sqrt{2gh}$ in that frame; back in the lab frame we add the prism's $v = \sqrt{2gh}$, so the total is $2\sqrt{2gh}$.